

Suggested problems: 1-11 odd

Problem 1: Suppose that a coin is flipped three times. Let X be the number of heads in any given sequence of tosses. What sort of structure is X ?

Problem 2: (Example 3.1.3) Recall the following problem from §1.5: A contractor is building an office complex and needs to plan for water and electricity demand. After consulting with prospective tenants and examining historical data, the contractor decides that the demand for electricity will range somewhere between 1 and 150 (in millions of kilowatt-hours per day) and water demand will be between 4 and 200 (in thousands of gallons per day). Recall that we defined a probability measure P for this experiment in terms of ratios of areas. Give several examples of random variables for this experiment.

Problem 3: (Example 3.1.4) Suppose that a coin is flipped ten times, and let X be the number of heads in a given sequence of tosses. What is the distribution of X ?

Problem 4: Referring back to Problem 2, what is the distribution of Z ?

Problem 5: Compute the probability function f of the random variable X in Problem 3.

Problem 6: Let $X \sim \text{Bin}(5, 0.75)$. Determine $P(X \leq 3)$.

Problem 7: Compute the probability function f of the random variable Z in Problem 2.